

Samira Carolina Oliva Madrigal

Erdős #4 ♦ Einstein #6 ♦ ORCID ♦ Google Scholar ♦ Personal Site ♦ GitHub ♦ LinkedIn ♦ scolivamadrigal@gmail.com

RELEVANT FORMATION

- Scientific Research & Cryptographic Engineering for Modern & Advanced Cryptography, TTL Logic Gate Design, Digital Design (Verilog), Computer Architecture and Design (MIPs, Verilog), Advanced Computer Design (Verilog), Application-Specific Design for Cryptosystems (Verilog/SystemVerilog), Microprocessor Design (Linux, C), Embedded-System Design (MIPs), Real-Time Embedded System Co-Design, Information Security, Algorithms and Data Structure Design (C/C++), Advanced Algorithm Design (C), System Software (C), Operating System Design (Linux, C), Compiler Design (Linux, C, x86, Lex), Software Engineering, Software Quality Assurance and Testing, Software Security Technologies, Computer Networks, Computer Network Design, Cryptography & Network Security, Network Architecture and Protocols, Network Programming and Applications, Advanced C Programming, C++ for C Programmers, Server-Side Web Programming, Assembly Language for IA 32 x86 Processors, UNIX/Linux, Shell Scripting, Numerical Analysis and Scientific Computing, Linear Algebra, Calculus-based Physics (Mechanics, E&M, Optics & Waves, & Particle (Special Relativity))

TECHNICAL SKILLS

Applied & Theoretical Cryptology, Expert in the Language of Cryptology, Implements across all domains (software, hardware, firmware (x86, MIPs), and co-designs), Scientific Research and Cryptographic Engineering in Modern and Advanced Cryptology, Internet TCP/IP Protocol Suite & Blockchain, ability to drive any area strongly

- **Background:** Pure & Applied Mathematics, Computer Science and Computer Engineering, Software Engineering, Systems Programming, Ethical Hacking, Secure Technologies (hardware/software bugs, exploits, etc.), Finite Field Arithmetic, Symmetric Primitive Design, Classical Primitives, Particle Physics, Advanced Cryptography including Post-Quantum Cryptography, Homomorphic Encryption, Cryptographic Proofs (e.g. Verifiable Computation, Verifiable Encryption, SNARK-friendly signatures), Privacy-Preserving Computation, and Quantum Algorithms

- **Realizations:** Constructions and Implementations, System Design, Implementing, Prototyping, and Testing (functional verification, unit testing, benchmarking, etc.)

- **Relevant Interests:** Privacy Preserving LLMs, Securing AI, Differential Privacy, Federated Learning, Quantum without Quantum Computation, ★ **Everything AI** ★, Natural Language Learning, NLP, Polyglot, Advanced Learning Algorithms (AI/ML)

- **Applied Math & Physics:** Field theory, proofs, problems and instances of problems on which crypto constructions are built, IFP, DL, ECDLP, NP problems, J-Invariant, SIS, SIVP, HPP, SVP, LWE, R-LWE, RSD, oil + vinegar, nonlinear multivariate systems of equations, NP-hard, applied linear algebra (e.g. code-based schemes and quantum computing), algebraic constructions, rings, modular multipliers, statistics, probability distributions, FFTs, calculus, differential equations, interference, parallelism

- **Cryptography & Protocols/Algorithms:** symmetric & asymmetric cryptography, KEX, *x.509*, PKI (RFC4949), CA, Kerberos, Layer 3 authentication and/or encryption, elliptic curve cryptography, sieving, OWFs, cryptanalysis, block cipher constructions and analysis, cryptographic hash functions, MACs, HMACs, digital signatures, PRFs, Montgomery, Blakely, BMM, interleaved multipliers, DES, 3DES, AES, RSA, DH, EC-DH, KECCAK, quantum algorithms (Grover, Shor, Simons), post-quantum cryptography, hash-based, lattice-based, code-based, multivariate-based, supersingular elliptic curve schemes, rank-based, consensus algorithms, Fiat-Shamir, Rainbow, McEliece, QC-McEliece, NTRU, CFS, SIDH, qRNG, parameter models (e.g. MOSS), bugs (Hardware, Firmware, & Software), Loquat, MPC-in-the Head & VOLE-in-the-Head paradigms, trusted or transparent ZKP constructions and applications, SumCheck Protocol, GKR, SNARK-friendly designs, Quantum without Quantum Computation, Experience with contributing to industry level cloud HSM (SDK, software algorithm specifications, key management, secure boot, confidential compute, feasibility studies, PQC prototyping, and co-designs for firmware updates)

- **Information Security:** confidentiality, authentication, integrity, secure coding, scanners, viruses, side-channel analysis, speculative execution, constant-time algorithms, gadgets, ROP/JOP, control-flow attacks, remote code execution, DDoS, oracles, **buffer overflows**, code injections, sniffers, backdoors, cloud, hypervisors, deep web, reconnaissance

- **Networking & Protocols/Algorithms:** topology setup, packet analysis, & testing of Internet protocols across all layers, signal processing, QAM-64, symbol/bit encoding schemes, error-correction, Media Access Control Schemes (e.g., CD-MAC, CA-MAC), ARP, NDP, Spanning Tree Protocol, IEEE 802.3, IEEE 802.11x, PPP, Tunneling, VNP, VLANs, QoS, IP (v4/v6), CIDR, RFC 1918, MPLS, Multicast, PIM (sparse, dense), IGMP (v4), MLP (v6), IPSec, NAPT, ICMP/v6, DNS, TLS, TCP, UDP, DIJKSTRA, OSPF, IS-IS, iBGP, eBGP, inter-AS routing, intra-AS routing, switching fabric, SDNs, control plane, data plane, Cloud (I/S/P/B as a Service), containers, microservices, sockets, Network OS (e.g., IOS XR) CLI, packet analysis, platform-agnostic (BSPs) system software

- **Digital & Analog Design:** Combinational & Sequential Circuits, Microarchitecture, Finite State Machine (FSM), Control Unit, Data-Path, Hierarchical Design, System-level Design, System Memory, FreeRTOS, Raspbian, microcontrollers with ARM cortex, LACP1769, LCPEXpresso, communication protocols (GPIO, UART, CAN, I2C, etc.), device drivers, Hardware Security, Probabilistic Models, Engineering Statistics

- **Programming: Expert in C (pointer-based language)**, OOP, C++, Java, **Verilog/SystemVerilog** HDLs, RISC (MIPs) and CISC (x86) ISAs, **Python**, Shell Scripting (bash, tcsh, bourne shell), Multithreading, Concurrency, Parallel Processing (with Python Ray), Virtualization, **SEI CERT C Coding Standard**, low-level code

- **Computer Science:** linear, non-linear, & dynamic **data structures** (e.g., trees, forests, and graphs), red-black, m-way trees, hash merkle trees, dynamic programming, complexity theory, space and time algorithmic complexity analysis, hardware analysis (CC count, cell count, critical path delay)

- **Industry Tools:** **Vivado**/ISE, **FGPAs** (Nexsys3, COM-1800, Virtex7), Digilent, **Xilinx Zynq UltraScale+ MPSoC ZCU102 Evaluation Kit** (cutting edge prototyping board used in doctoral research, industry, and government), **Xcode/gcc/NASM/PyCharm**/Eclipse/Visual Studio/MIPs Assembler/MASM, MATLAB, Pytest, TextFSM, Wireshark, routers (ASR9K, NCSxx), switches, line cards, Spirent/Ixia traffic generators, testbed setup, Jenkins, VMs, OS: MacOS, Windows, UNIX/Linux distros (e.g., Fedora, Debian, Ubuntu, CentOS)

- **Public Learning Tools:** Cisco Delcloud, Amazon VPC, GNS3, IBM Quantum/Qiskit, virtual classrooms

- **Familiar with:** **Rust**, PKCS # 11, Open Source Projects (e.g., OQS), Go, DAPPs in Solidity, **kernel programming**, kernel modules, platform firmware, trusted hardware, ARM TrustZone, EFI, UEFI, Docker & Kubernetes, building a container from scratch, FIPS-140-3 and related ISO standards, PIN cracking, Payment Card Industry (PCI) Security Standards (e.g., Crypto Key Blocks), Quantum Algorithms & Protocols

(Qiskit & Jupyter Notebook), LinuxBIOS and patching OpenSSL source code (assembly cryptographic code, BN, Envelope Encryption, and API), Homomorphic Encryption (e.g. Fan-Vercauteren, RLWE), Side-channels (e.g., table lookups and modular reductions), ensuring constant time algorithms, NIST PQC 3rd Round Finalist's documentation and implementations in C, zk-Proofs (from QAPs and EC pairings with HE), zk-SNARKs (e.g., Pincocchio & Aurora), zk trusted setup with Multi-Party Computation (e.g., Zcash), Number Theoretic Transforms

KEY FACETS

- Self-starter, likes to benchmark work against state-of-the-art, fast learner, works excellent in group or individual

EDUCATION

2025

Télécom Paris / l'Institut Polytechnique de Paris. Palaiseau, France

PhD Computer Science: Cryptology

Thesis: Symmetric-key-based Post-Quantum Advanced Cryptography ← click

Fully funded doctoral three year program with long stay talent researcher French VISA at the French grande école as part of:

Le Laboratoire de Traitement et Communication de l'Information (LTCI) lab within L'équipe Cybersécurité et Cryptographie [C²] group

Objectives: considered opportunity to develop strongly abroad in ideal setting to contribute individual ideas as an independent researcher and interesting collaborations, write a good thesis, and become fluent in French. Rigorous study and construction of post-quantum secure cryptographic proofs from PIOP from (ideally) symmetric-key-based primitives with special interest for instances for SNARK-friendly signatures and verifiable encryption and contributions to other related advanced primitives as standalone areas, such as fully homomorphic encryption, post-quantum cryptography, and abstract algebra. Interests for contributions from quantum proofs sans quantum computation and continued work for bootstrapping for CKKS-RNS and verifiable computation.

Thesis concerned areas within my strengths: state-of-the-art in cryptographic proofs (PIOP+PCS constructions), abstract algebra, homomorphic and fully homomorphic encryption, distributed protocols, constructions from simulated MPC protocols, symmetric primitives (OWFs, PRFs, etc.), post-quantum cryptography, and others. Instances of focus: SNARK-friendly signatures, verifiable encryption (e.g. for AES, CKKS-RNS, etc.), using encryption as a commitment, anonymous credentials, and others. Incredibly vast and interesting applications from general transaction validation to path authentication in the notorious Border Gateway Protocol (BGP).

Successfully conducted research for areas of interest ~ 912 hours ★ reviewed dozens of articles ★ applied rigorous treatment of material ★ most recent work verified two major improvements to works of interest ★ proposed various novel improvements which qualified for standalone publications and applications to improve distinct problems ★ contributed several ideas for original and improved work in the area of cryptographic proofs (zkSNARKs and zkSTARKs) and homomorphic encryption. Offered fully funded program for [Fall school on Geometry in Cryptography and Communication](#) October 2025 at UiT The Arctic University of Norway, Tromsø and invited to present RF-FIKO to mathematical community ★ successfully defended thesis on 29 September 2025 ★ successfully completed all planned work towards formation: 6 hours of Scientific Research Ethics and 37-38 hours of Advanced French with 100% completion and 99.30% GPA to validate 20h of a foreign language ★ participated in several conferences some which took place in Paris and others in Palaiseau; 2 reading group meetings at INRIA.

★ Targeted the 40h non-thesis related scientific formation of the 100h formation with geometry of polynomials and side-channel attacks in hardware with state-of-the-art prototyping boards.

★ **Early Research Foundations:** My research career began with high-impact, independent scientific inquiry with Koç Lab under world-renowned cryptographer Prof. Dr. Çetin Kaya Koç. This foundation included research collaborations tied to Villanova University and an international research collaboration with Nanjing University, which resulted in my very first peer-reviewed publication. This was in parallel with **Elite Industry Seniority:** as a former Senior Engineer in Cryptology at Marvell where I led platform-agnostic software-hardware co-design feasibility studies for advanced algorithms and was the lead developer and cryptographic firmware contributor for production-level Post-Quantum Cryptography (PQC) feature, leading two remote teams.

In Fall 2024, I faced a strategic decision between two fully funded doctoral opportunities: continuing my established work at Villanova University or accepting a highly competitive, three-year program at Télécom Paris (l'Institut Polytechnique de Paris). Having already spent two years producing research with Dr. Koç and Dr. Xie, I knew I possessed the technical autonomy to drive advanced scientific inquiry independently. I chose the fellowship in Paris—a global epicenter for mathematical and cryptologic excellence—to intentionally pressure-test my scholarship in a new international environment.

During my time at Télécom Paris/IPP, I provided immediate direction, explained material in detail, quickly verified results in paper and implemented, and verified computations for the thesis director. **Executive R&D Leadership:** recognizing that the rapid velocity and applied nature of my research outpaced traditional academic timelines, I strategically chose to conclude the fellowship track early to translate my independent research work into an autonomous commercial paradigm. I continued building Cryptologia, LLC, an independent cryptologic R&D laboratory and consultancy firm where I serve as Chief Cryptographer alongside Dr. Çetin Kaya Koç. **Top-Tier Scientific Output:** My independent research trajectory routinely exceeds the milestones of a standard doctoral track. Operating as an autonomous principal investigator, I focus on publishing peer-reviewed breakthroughs in premier global venues—such as CASA and SMALL—demonstrating a scientific output that surpasses traditional institutional benchmarks for excellence. **I focus on building elite, long-term relationships with academia leveraging my position as an independent principal investigator to bridge the gap between industry and academia.** Cryptologia, LLC targets high-impact, collaborative research initiatives and open up opportunities to guide student R&D with ultimate objective to deliver the level of groundbreaking institutional impact and societal contribution that merits future **honorary doctoral recognition.**

2025	l'Institut Polytechnique de Paris. Paris, France 38h to validate 20h: Advanced French Certificate with 99.30% GPA & 6h Scientific Research Ethics - Certificate
2025	De Componendis Cifris, Milano, Italy / Università di Trento, Trento, Italy Course Attendance Certificate - De Cifris Trends in Cryptographic Protocols - The French Magisterium (2024) Attended and passed exam for Trends24 from Associazione De Componendis Cifris and Università di Trento, Department of Mathematics. Program consisted of 20 lectures in The Almost Perfect Nonlinearity of Substitution Boxes and its Consequences, Algebraic aspects in designing cryptographic functions in symmetric cryptography, Lightweight symmetric primitives, Symmetric Techniques for Advanced Protocols: Design Strategies, and Cryptanalysis, The Transition to Post-Quantum Cryptography, Lattice-based Cryptography (I) & (II), Hardness of the Module Learning With Errors Problem, Diving into Multivariate Cryptography, The Code Equivalence Problem and its Applications to Cryptography, Code-based Cryptography, The Alekhnovich cryptosystem: code-based cryptography with security proofs, Multi-Party Computation in the Head: Techniques and Applications, Polynomial System Solving and Application to Algebraic Cryptanalysis, Tools for cryptanalysis of symmetric primitives, Side-Channel Attacks and Masking Countermeasures, Fully Homomorphic Encryption, Hybrid Homomorphic Encryption, Cryptography for anonymity and accountability, Anonymous Tokens from leading French Cryptographers.
2022 - 2024	Villanova University, Villanova, PA Research and bootstrapping for CKKS-RNS with hardware draft in SystemVerilog based on state-of-the-art work from the Lattigo library implementation in Golang. Collaborated for two excellent publications in 2024 & 2025 (CASA and SMALL). Proposal for 2 year remote formal PhD in ZKP & hardware-based bootstrapping.
2024	De Componendis Cifris, Milano, Italy / Università di Trento, Trento, Italy Course Attendance Certificate - De Cifris Trends in Cryptographic Protocols 2023 Attended and passed exam for Trends23 from Associazione De Componendis Cifris and Università di Trento, Department of Mathematics. Program consisted of lectures in Security and Composition of Cryptographic Protocols, Zero-Knowledge Protocols, Sigma protocols, Vector commitments, Fully Homomorphic Encryption, Threshold Cryptographic Protocols, Private Set Intersection, Hierarchical Key assignment, Protocols for Peer Rating Systems, and Advanced Cryptography in E-Voting from leading Cryptographers.
2021	University of Buenos Aires (virtual ECI34), Argentina Certificate of Achievement - Quantum Random Number Generators .
2018 - 2019	San José State University, San José, CA M.Sc. Computer Engineering with 3.571 GPA Double Specialization: Networking Systems & Secure Systems Thesis: <i>Reduction-free Multiplication in $GF(2^n)$ Applicable to Modern and PQC schemes</i>
2013 - 2017	San José State University, San José, CA B.Sc. Computer Engineering, Minor Computer Science with 3.362 GPA Senior Project: <i>FPGA-based Blockchain Accelerator for Ethereum Proof-of-Work</i>
2010 - 2013	De Anza College, Cupertino, CA A.A. Systems Programming with 3.46 GPA; French & Italian Studies with 4.0 GPA

PUBLICATIONS

Samira Carolina Oliva Madrigal. Improvements to Field Membership Calculus in $GF(2^n)$ from the Null Law with Applications to Verifiable Computation *International Workshop on Arithmetic of Finite Fields (WAIFI)*. [Submitted March 2026](#).

P. He, S. C. Oliva Madrigal, Ç. K. Koç, T. Bao, and J. Xie. SMALL: Scalable Matrix OriginAteD Large Integer PoLynomial Multiplication Accelerator for Lattice-based Post-Quantum Cryptography. *International Workshop on Arithmetic of Finite Fields (WAIFI)* Ottawa, Canada, to appear, June 10-12, 2024., [Publication](#)

P. He, S. C. Oliva Madrigal, Ç. K. Koç, T. Bao, and J. Xie. CASA: A Compact and Scalable Accelerator for Approximate Homomorphic Encryption. *International Association for Cryptologic Research (IACR) Transactions on Cryptographic Hardware and Embedded Systems*, Volume 2024, No. 2, to appear, 2024., [Publication](#)

S. C. Oliva Madrigal, G. Saldamlı, C. Li, Y. Geng, T. Jing, Z. Wang, and Ç. K. Koç. 2023. Reduction-Free Multiplication for Finite Fields and Polynomial Rings. In *Arithmetic of Finite Fields: 9th International Workshop, WAIFI 2022, Chengdu, China, August 29 - September 2, 2022, Revised Selected Papers*. SpringerVerlag, Berlin, Heidelberg, 53–78. [Publication](#)

Samira Carolina Oliva Madrigal 2019. Reduction-free Multiplication in $GF(2^n)$ Applicable to Modern and Post-quantum Cryptographic Schemes. *San Jose State University. Computer Engineering*. [Publication](#)

PRESENTATIONS

Presented recent work at WAIFI 2024: Scalable Matrix OriginAteD Large Integer PoLynomial Multiplication Accelerator for Lattice-based Post-Quantum Cryptography (SMALL). *International Workshop on Arithmetic of Finite Fields (WAIFI)* Ottawa, Canada, June 10-12, 2024.

Presented paper on behalf of the authors, previous collaborators: Chen Li, Suwen Song, Jing Tian, Zhongfeng Wang, and Çetin Kaya Koç. An efficient hardware design for fast implementation of HQC. *IEEE 36th International System-on-Chip Conference (SOCC)*, Santa Clara, California, pages 1-6, September 5-8, 2023.

Presented at WAIFI 2022 (virtual conference): Reduction-free Multiplication in $GF(2^n)$ Applicable to Modern and Post-quantum Cryptographic Scheme. *International Workshop on Arithmetic of Finite Fields (WAIFI)* Chengdu, China, August 29-September 2, 2022

LICENSES & CERTIFICATIONS

CRYPTOLOGIA, LLC - Business License

Business Licensed from the City of Campbell, registered entity with the State of California, and registered business across numerous governing bodies including (Beneficial Ownership Information Report (BOIR), Financial Crimes Enforcement Network (FINCEN), etc.) to conduct business remotely.

Cryptologic R&D and Consultancy Services Lab.

Business to Business Services (B2B) currently serving Post-Quantum Cryptography and AI adoption (1099 or W-2)

University of Maryland Hardware Security, IP

edureka! Generative AI for Data Privacy & Protection, grade: 100%

Stanford University Machine Learning Specialization: 90 hours, grade: 100%

Supervised Machine Learning Specialization: Unsupervised Learning, Recommenders, Reinforcement Learning, grade: 100%

Supervised Machine Learning Specialization: Advanced Learning Algorithms, grade: 100%

Supervised Machine Learning Specialization: Regression and Classification, grade: 100%

Anthropic Claude Code in Action, grade: 100%

Scrimba OpenAI Codex, grade: 100%

EDUCBA Advanced Smart Contract Development, grade: 100%

INSEAD Web3 and Blockchain, grade: 100%

Google Leadership & Influencing, grade: 100%

Google Google AI Specialization, grade: 100%

RESEARCH EXPERIENCE

- Active Privacy Preserving LLMs, Cryptographic Proofs, Abstract Algebra, MPC Protocols, Post-Quantum Cryptography, FHE ((bootstrapping [Lattigo & SystemVerilog]), hardware, embedded
- 2022 ZK-Proofs, SNARKs, Multi-Party Computation, Fully Homomorphic Encryption, Proofs → Algorithms → Implementation
- 2021 Quantum Computing & qRNG; BaaS: Hyperledger Forks, Quantum-Securing the Blockchain, Programmable Blockchain SDKs, token-agnostic bartering, & variants
- 2019 San José State University, San José, CA
NSF Post-Quantum Cryptography Proposal
- 2019 San José State University, San José, CA
Modular Multiplication in $GF(2^n)$
- 2016 San José State University, San José, CA
Blockchain Industry & Distributed Applications

RELEVANT PROFESSIONAL EXPERIENCE

- 2025 - present **Cryptologia, LLC**
Founder & Chief Cryptographer
Remote-first based AI-centric Cryptologic Scientific Research & Development and Consultancy Services Laboratory
- 2022 - 2024 **Expert Research Team with Koç Lab & Private R2 Research Villanova University**
Scientific Researcher & Engineer
Research in FHE and PQC (Remote)
 - Collaboration with small group of Cryptography and Engineering experts and PhD students
 - Focus is optimized arithmetic for FHE and PhD work
 - Published in IARC for leading RNS-CKKS work (2024)
 - Initial draft for bootstrapping in hardware for RNS-CKKS based on Lattigo library
 - Published & presented work for WAIFI 2024 (competitive selection) for optimized arithmetic in Lattice-based PQC with applications to HE (2024)
- 2022 - 2024 **Marvell, Santa Clara, CA**
Senior Engineer, Cryptology
Applied Cryptography and Development Work in Post-Quantum Cryptography
 - algorithm breakdown & analysis and Protocols
 - OpenSSL, TLS, FIPS 203, 204, 205, Falcon, Hash-based Signature Schemes (HBS) e.g., HSS & LMS
 - underlying mechanism based on Fiat-Shamir paradigm and zero knowledge proofs
 - Cryptographic firmware in C; Interfacing with hardware; Software-Hardware Co-Design mapping in Verilog/SystemVerilog and interfacing and mapping for cryptographic core(s) and microcode mapping
 - lead cryptographer and developer for PQC; trained and collaborated with two teams
 - Platform-agnostic proof of concept solution for acceleration with software-hardware co-design
 - vast span across cryptographic engineering work: algorithm assessment and recommendations, cryptographic software, firmware, hardware, research, prototyping, library patching, software requirements specifications, design documents, product mapping, production level development, benchmarking, gtest, unit testing, scripting, end-to-end testing, algorithm optimizations
- 2021 - 2022 **Startup**
Research Scientist for architecture and development of quantum-secure cryptographic protocols for p2p application.
- Fall 2019 **San José State University, San José, CA**
Instructional Student Assistant for graduate course in Cryptography and Network Security.
 - Course covered Galois Field Arithmetic, Public-key & Symmetric-key Cryptosystems, Digital Signatures, Authentication, Kerberos, PKIs, Certificates, and L5/3 Security Protocols.

2017 - 2018 - Prepared review notes for students and graded homework assignments, quizzes, and exams.

Cisco Systems, Inc., Milpitas, CA

Software Engineer for feature testing and automation of next-generation Service Provider.

- Automated testing of network operating system protocols on different router platforms.
- Unit testing, code review, bug resolution with developers, regression testing, and mentored a remote colleague.
- Technology Stack: Routers, Switches, Traffic Generators, Testbed setup, VMs, GitHub, Jenkins, Linux, Python, and Shell Scripting.

RELEVANT ACADEMIC PROJECTS & INDEPENDENT RESEARCH

2025	Research in Cryptographic Proofs, Verifiable Computation, FHE, & Improving works with Modern and Advanced Cryptography
2024	Bootstrapping for CKKS-RNS with SpareKeys Hardware Draft
2022	Fundamental Zero-Knowledge Protocols with RSA, Schnorr, and discrete log zk-SNARK
2022	Partially Homomorphic Encryption with RSA
2021	AES Software Implementation in C based on FIPS-197
2021	KECCAK Software Implementation in C based on FIPS-202
2021	RSA Software Implementation in C using OpenSSL BN data structure
2021	RSA Software Implementation in C using OpenSSL Envelope Encryption API
2021	$GF(2^n)$ Multiplication in x86 NASM assembly (32/64-bit)
2019	(Group) Steganography Python Application with TLS (OpenSSL, virtual datastore, & sockets)
2019	Public-Key Infrastructure Application using x.509 certificates
2019	Index-Calculus Research Project
2016	(Team) Hardware Implementation of KECCAK based on FIPS-202
2016	AES Hardware Implementation in SystemVerilog based on FIPS-197
2015	(Team) 32-bit Pipelined MIPS Processor (Verilog)
2014	Crypto Workhorse: Block-Cipher Study with Focus on AES and DES

AWARDS & HONORS

2026	Featured in Influential Women 2026
2024	Marvell Recognition: 7 awards from post-quantum cryptography development team members, including technical leader and director.
2023	Marvell - Selected for company branding and story telling feature: Ownership
2023	Marvell CEO - Game Changer Engineer Award for contributions to Post-Quantum Cryptography (PQC)
2023	Marvell VP Award for PQC
2022	Director & Team Recognition for rigor and innovation in PQC
2019	Best Homework for graduate course in network programming and applications
2017	Cisco You Inspire 2 Award - Energetic engineer who takes up lab activities
2017	Dean's Scholar - 55th annual Honor's Convocation for GPA of 3.64+ for 2+ contiguous semesters

LANGUAGES

- Polyglot fluent in Spanish, English & Italian; Professional working French; Beginner Russian, Portuguese;

ACTIVITIES

- Active Supporter of Rome's Call for [ALGORETHICS](#) & Discerning AI
- IACR ([Crypto 2020](#) & [2021](#) and [PKC 2022](#) Conferences), [EITCI](#), [Attended California assembly member AI Seminar \(March 2026\)](#), OpenAI Academy (on going), Industry Rockstar AI Consultant SUMMIT (April 29, 2026)
- Volunteering at [St. Lucy Catholic Parish](#), walks in nature & working out & running & reading & karaoke & foreign languages
- [FHE research with academic group since July 2022 - 2024 \(hardware focused\) and published work on leading cryptographic journal; transition to AI Security](#)
- [Participated in De Cifris Trends in Cryptographic Protocols 2023 & The French Magisterium 2024](#)
- [Participated in several special cryptography conferences in Paris & Palaiseau, France \(2025\) including - CRYPTO Day Conferences @ Télécom Paris, SecureCompute 2025 @ ENS Paris, Special Crypto Day at IHP, SEIDO 2025](#)